
Course Outline and Evaluation Policy
SC471: Algorithmic Graph Theory during Autumn 2025-26
Course Instructor: V. Sunitha

1. Title: Algorithmic Graph Theory

2. Credit Structure (L-T-P-Cr): 3 0 2 4

3. Course Code: SC471

4. Semester: Sem V, Sem VII of B.Tech. (MnC) / Sem VII of B.Tech. (ICT, ICT-CS) / Sem I of M.Tech. (ICT)

5. Category: MnC Elective / Science Elective / General Elective (Maths)

6. Prerequisite: **Exposure to logical reasoning and proof techniques, ability to write neat computer programs, willingness to actively participate in classroom discussions.**

7. Objective/Motivation: Graphs have a wide range of applications across various fields. They have proven to be an invaluable tool in establishing results in numerous areas of mathematics and, by extension, in engineering. Many graph-theoretic proofs are constructive, leading directly to the development of algorithms for solving diverse problems and puzzles. An engineer with a solid understanding of graph theory and its algorithms is better equipped to address and solve a broad array of complex challenges efficiently.

8. Outcomes: The student should be able to model and solve real-world problems algorithmically using graph-theoretic techniques, while also providing a proof of algorithm correctness. In cases where efficient algorithms are unavailable, the student should be capable of developing practical, effective alternatives. Furthermore, the student should experience a notable improvement in both analytical and programming skills.

9. Topics (not including definitions, examples and other trivialities):

- **A collection of results with emphasis on the following fundamental theorems and algorithms on graphs:**

<i>Theorems</i>	<i>Algorithms</i>
Handshaking lemmas, Havel-Hakimi theorem, Erdos and Gallai theorem, The matrix-tree theorem, Cayley-Kirchoff theorem, Menger's theorem(s), Characterization of Eulerian graphs, Dirac's theorem, Ore's theorem, Chvatal's theorem,	Graph traversal algorithm(s), Shortest path algorithm(s), Warshall's algorithm, Prufer sequence and labeled tree construction, Minimum spanning tree algorithm(s), Fleury's algorithm, Colouring algorithm(s), Planarity testing algorithm(s), Finding strong components, Ranking in

Hall's matching condition, Konig-Egervary theorem, Tutte's theorem, Brook's theorem, Vizing's theorem, Shannon's theorem, Robertson-Seymour theorem, Kuratowski's theorem, Five colour theorem	tournaments. In case of NP-complete problems, polynomial algorithms available for special cases will be discussed.
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10. References: Vast resources on graph theory are available today and one has an easy access to many of these either online or in our RC. The following collection is a good starting point.

- A. Gibbons, Algorithmic Graph Theory, Cambridge University Press.
- S. Even, Graph Algorithms, Computer Science Press.
- D.B. West, Introduction to Graph Theory (2nd edition), Prentice Hall of India.
- J.A. Bondy and U.S.R. Murty, Graph Theory with Applications, MacMillan Press.
- G. Agnarsson and R. Greenlaw, Graph Theory: Modelling, Applications and Algorithms, Pearson Prentice Hall.
- R. Diestel, Graph Theory (3rd edition), Graduate Texts in Mathematics Series, Springer.
- J. B-Jensen and G. Gutin, Digraphs: Theory, Algorithms and Applications, Springer-Verlag.

11. Lab sessions: Mostly unsupervised and will involve writing graph algorithms, analysis as well as proof of correctness of these algorithms, and optionally computer programs for the algorithms. A short introduction will be given for each lab assignment and the students are expected to work independently on the assignment. Access to Institute's Computing facilities for assignments requiring the same shall be provided as per the lab time-table for the course.

12. Assessment: The assessment will consist of the following components, each contributing 25% to the final grade:

1. Exam-1
2. Exam-2
3. Exam-3
4. Assignments and Quizzes

Plagiarism of any kind will result in an 'F' grade.

- **Exam-1** will take place during the in-semester 1 examinations, as per the Autumn 2025-26 Academic Calendar, and will cover topics taught up until that point.
- **Exam-2** will be held during the in-semester 2 examinations and will cover topics taught after Exam-1.
- **Exam-3** will take place during the end-of-semester examinations, covering topics taught after Exam-2.

If the number of students enrolled in the course permits, a project may be introduced as part of the evaluation. In that case, the weightage for all components will be adjusted equally.

13. Placement, work load, illness, and other absence: **There shall be no re-exam or make-up exam. Any assignment not submitted on time and all unattended quizzes shall be awarded zero marks.**

14. Grading Policy: **Absolute grading as per the following (tentative) scheme**

Range of overall score (in %, left inclusive)	85 - 100	70 - 85	60 - 70	50 - 60	40 – 50	35 – 40	30 – 35	25 - 30	0 - 25
Grade	AA	AB	BB	BC	CC	CD	DD	DE	F